

System Parameters (defined by hardware)			Profiles (usage of each component mode - defined by software and usage)			
from the datasheets			"off"	"sensing"	"interactive"	
Supply (V)	3.3					
Processor		Current Consumption (mA)				
Active (modem sleep)	89 mW	27	0%	10%	90%	
Idle	0 mW	0.13	0%	85%	10%	
Sleep	0 mW	0.005	100%	5%	0%	
LED						
On	1 mW	0.17	0%	100%	100%	
Sensor						
On	69 mW	21	0%	100%	0%	
Idle	0 mW	0.009	0%	0%	100%	
Off	0 mW	0.007	100%	0%	0%	
Display						
On	0 mW	0	0%	0%	0%	
Off (leakage)	0 mW	0	0%	0%	0%	
Radio						
Data Rate	400 bps					
Standby Power	3 mW					
TX Power	85 mW					
RX Power	85 mW					
Battery			20	2	2	hours/day typical usage
Capacity	1892 mAh					
Nominal Voltage	4 V					
Regulator Efficiency	85%					

REFLECTIONS : WHAT DID YOU LEARN FROM ANALYZING YOUR POWER. TALK ABOUT SOME POTENTIAL TRADEOFFS.

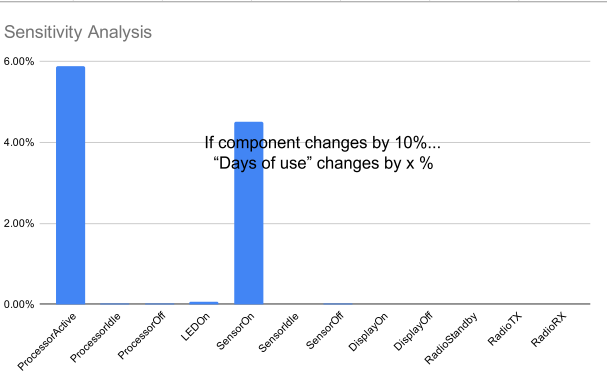
I learned that RF mode and Work modes are two different modes. So I was initially analyzing the ESP32C3 datasheet incorrectly. For my device my "active mode" is actually Modem Sleep mode, because it's basically Idle mode until the device triggers the sensor.

These parameters are really rough estimates as I'm doing this last minute with out instructor help. But my prediction matches "Hours of Use" where the device will be off or idle more than it's being used and therefore the battery will last for a while. Neopixel current consumption is used for LED. I used the max use-case scenario which is bright white light just in case higher battery capacity is needed, this can provide more battery power than what is needed. Of course this is still a tradeoff, because if I'm trying to be price efficient more than power efficient I'm analyzing battery power more than what is required.

How did you determine your "days of use" metric? I gave most of my fraction into Idle modes and off modes because it'll be a short time in sensing mode in my use case scenario which is taking measurements from point A to point B.

What do you think is the optimum size for the battery in your device? Anything that fits 3V3 battery

What hardware/software/cost/effort tradeoffs could you make to improve the user experience?
I don't need shifters or capacitor pairs just pull up resistors for sensing device. That is sufficient enough to use the sensor



Total power in profile (mw)		Maximum Time
"off"	0.0396 mW	150261.1 hours
"sensing"	79.136475 mW	75.2 hours
"interactive"	80.8236 mW	73.6 hours
Effective Battery Capacity		
5950.34 mW*h		
Days of Use	18.55 days	
Hours of Use	445.28 hours	